

DOUBLES FOR MONOIDAL CATEGORIES

Dedicated to Walter Tholen on his 60th birthday

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ABSTRACT. In a recent paper, Daisuke Tambara defined two-sided actions on an endomodule (= endodistributor) of a monoidal $\mathcal{V}$ -category $\mathcal{A}$. When $\mathcal{A}$ is autonomous (= rigid = compact), he showed that the $\mathcal{V}$ -category (that we call $\text{Tamb}(\mathcal{A})$) of so-equipped endomodules (that we call Tambara modules) is equivalent to the monoidal centre $\mathcal{Z}[\mathcal{A}, \mathcal{V}]$ of the convolution monoidal $\mathcal{V}$ -category $[\mathcal{A}, \mathcal{V}]$. Our paper extends these ideas somewhat. For general $\mathcal{A}$, we construct a promonoidal $\mathcal{V}$ -category $\mathcal{DA}$ (which we suggest should be called the double of $\mathcal{A}$) with an equivalence $[\mathcal{DA}, \mathcal{V}] \simeq \text{Tamb}(\mathcal{A})$. When $\mathcal{A}$ is closed, we define strong (respectively, left strong) Tambara modules and show that these constitute a $\mathcal{V}$ -category $\text{Tamb}_s(\mathcal{A})$ (respectively, $\text{Tamb}_{ls}(\mathcal{A})$) which is equivalent to the centre (respectively, lax centre) of $[\mathcal{A}, \mathcal{V}]$. We construct localizations $\mathcal{D}_s\mathcal{A}$ and $\mathcal{D}_{ls}\mathcal{A}$ of $\mathcal{DA}$ such that there are equivalences $\text{Tamb}_s(\mathcal{A}) \simeq [\mathcal{D}_s\mathcal{A}, \mathcal{V}]$ and $\text{Tamb}_{ls}(\mathcal{A}) \simeq [\mathcal{D}_{ls}\mathcal{A}, \mathcal{V}]$. When $\mathcal{A}$ is autonomous, every Tambara module is strong; this implies an equivalence $\mathcal{Z}[\mathcal{A}, \mathcal{V}] \simeq [\mathcal{DA}, \mathcal{V}]$.

1. Introduction

For $\mathcal{V}$ -categories $\mathcal{A}$ and $\mathcal{B}$, a *module* $T : \mathcal{A} \multimap \mathcal{B}$ (also called “bimodule”, “profunctor”, and “distributor”) is a $\mathcal{V}$ -functor $T : \mathcal{B}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{V}$. For a monoidal $\mathcal{V}$ -category $\mathcal{A}$, Tambara [Tam06] defined two-sided actions α of $\mathcal{A}$ on an endomodule $T : \mathcal{A} \multimap \mathcal{A}$. When $\mathcal{A}$ is autonomous (also called “rigid” or “compact”) he showed that the $\mathcal{V}$ -category $\text{Tamb}(\mathcal{A})$ of Tambara modules (T, α) is equivalent to the monoidal centre $\mathcal{Z}[\mathcal{A}, \mathcal{V}]$ of the convolution monoidal $\mathcal{V}$ -category $[\mathcal{A}, \mathcal{V}]$.

Our paper extends these ideas in four ways:

1. our base monoidal category $\mathcal{V}$ is quite general (as in [Kel82]) not just vector spaces;
2. our results are mainly for a closed monoidal $\mathcal{V}$ -category $\mathcal{A}$, generalizing the autonomous case;
3. we show the connection with the lax centre as well as the centre; and,
4. we introduce the double $\mathcal{DA}$ of a monoidal $\mathcal{V}$ -category $\mathcal{A}$ and some localizations of it, and relate these to Tambara modules.

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Our principal goal is to give conditions under which the centre and lax centre of a $\mathcal{V}$ -valued $\mathcal{V}$ -functor monoidal $\mathcal{V}$ -category is again such. Some results in this direction can be found in [DS07].

For general monoidal $\mathcal{A}$, we construct a promonoidal $\mathcal{V}$ -category $\mathcal{D}\mathcal{A}$ with an equivalence $[\mathcal{D}\mathcal{A}, \mathcal{V}] \simeq \text{Tamb}(\mathcal{A})$. When $\mathcal{A}$ is closed, we define when a Tambara module is (left) strong and show that these constitute a $\mathcal{V}$ -category $(\text{Tamb}_{ls}(\mathcal{A}))$ $\text{Tamb}_s(\mathcal{A})$ which is equivalent to the (lax) centre of $[\mathcal{A}, \mathcal{V}]$. We construct localizations $\mathcal{D}_s\mathcal{A}$ and $\mathcal{D}_{ls}\mathcal{A}$ of $\mathcal{D}\mathcal{A}$ such that there are equivalences $\text{Tamb}_s(\mathcal{A}) \simeq [\mathcal{D}_s\mathcal{A}, \mathcal{V}]$ and $\text{Tamb}_{ls}(\mathcal{A}) \simeq [\mathcal{D}_{ls}\mathcal{A}, \mathcal{V}]$. When $\mathcal{A}$ is autonomous, every Tambara module is strong, which implies an equivalence $\mathcal{Z}[\mathcal{A}, \mathcal{V}] \simeq [\mathcal{D}\mathcal{A}, \mathcal{V}]$. These results should be compared with those of [DS07] where the lax centre of $[\mathcal{A}, \mathcal{V}]$ is shown generally to be a full sub- $\mathcal{V}$ -category of a functor $\mathcal{V}$ -category $[\mathcal{A}_M, \mathcal{V}]$ which also becomes an equivalence $\mathcal{Z}[\mathcal{A}, \mathcal{V}] \simeq [\mathcal{A}_M, \mathcal{V}]$ when $\mathcal{A}$ is autonomous.

As we were completing this paper, Ignacio Lopez Franco sent us his preprint [LF07] which has some results in common with ours. As an example for $\mathcal{V}$ -modules of his general constructions on pseudomonoids, he is also led to what we call the double monad.

2. Centres and convolution

We work with categories enriched in a base monoidal category $\mathcal{V}$ as used by Kelly [Kel82]. It is symmetric, closed, complete and cocomplete.

Let $\mathcal{A}$ denote a closed monoidal $\mathcal{V}$ -category. We denote the tensor product by $A \otimes B$ and the unit by I in the hope that this will cause no confusion with the same symbols used for the base $\mathcal{V}$ itself. We have $\mathcal{V}$ -natural isomorphisms

$$\mathcal{A}(A, {}^B C) \cong \mathcal{A}(A \otimes B, C) \cong \mathcal{A}(B, C^A)$$

defined by evaluation and coevaluation morphisms

$$e_l : {}^B C \otimes B \longrightarrow C, \quad d_l : A \longrightarrow {}^B(A \otimes B), \quad e_r : A \otimes C^A \longrightarrow C \quad \text{and} \quad d_r : B \longrightarrow (A \otimes B)^A.$$

Consequently, there are canonical isomorphisms

$${}^{A \otimes B} C \cong {}^A({}^B C), \quad C^{A \otimes B} \cong (C^A)^B, \quad ({}^B C)^A \cong {}^B(C^A) \quad \text{and} \quad {}^I C \cong C \cong C^I$$

which we write as if they were identifications just as we do with the associativity and unit isomorphisms. We also write ${}^B C^A$ for ${}^B(C^A)$.

The Day convolution monoidal structure [Day70] on the $\mathcal{V}$ -category $[\mathcal{A}, \mathcal{V}]$ of $\mathcal{V}$ -functors from $\mathcal{A}$ to $\mathcal{V}$ consists of the tensor product $F * G$ and unit J defined by

$$\begin{aligned} (F * G)A &= \int^{U, V} \mathcal{A}(U \otimes V, A) \otimes FU \otimes GV \\ &\cong \int^V F({}^V A) \otimes GV \\ &\cong \int^U FU \otimes G({}^U A) \end{aligned}$$

and

$$JA = \mathcal{A}(I, A).$$

In particular,

$$(F * \mathcal{A}(A, -))B \cong F({}^AB) \quad \text{and} \quad (\mathcal{A}(A, -) * G)B \cong G(B^A).$$

The centre of a monoidal category was defined in [JS91] and the lax centre was defined, for example, in [DPS07]. Since the representables are dense in $[\mathcal{A}, \mathcal{V}]$, an object of the *lax centre* $\mathcal{Z}_l[\mathcal{A}, \mathcal{V}]$ of $[\mathcal{A}, \mathcal{V}]$ is a pair (F, θ) consisting of $F \in [\mathcal{A}, \mathcal{V}]$ and a $\mathcal{V}$ -natural family θ of morphisms

$$\theta_{A,B} : F({}^AB) \longrightarrow F(B^A)$$

such that the diagrams

$$\begin{array}{ccc} F({}^{A \otimes B}C) & \xrightarrow{\theta_{A \otimes B, C}} & F({}^{C^{A \otimes B}}) \\ \downarrow = & & \uparrow = \\ F({}^A({}^BC)) & & F({}^{((C^A)^B)}) \\ \searrow \theta_{A, BC} & & \nearrow \theta_{B, CA} \\ & F({}^{BC^A}) & \end{array} \quad \text{and} \quad \begin{array}{ccc} F({}^IA) & \xrightarrow{\theta_{I, A}} & F({}^{A^I}) \\ \searrow = & & \nearrow = \\ & FA & \end{array}$$

commute. The hom object $\mathcal{Z}_l[\mathcal{A}, \mathcal{V}]((F, \theta), (G, \phi))$ is defined to be the equalizer of two obvious morphisms out of $[\mathcal{A}, \mathcal{V}](F, G)$. The *centre* $\mathcal{Z}[\mathcal{A}, \mathcal{V}]$ of $[\mathcal{A}, \mathcal{V}]$ is the full sub- $\mathcal{V}$ -category of $\mathcal{Z}_l[\mathcal{A}, \mathcal{V}]$ consisting of those objects (F, θ) with θ invertible.

3. Tambara modules

Let $\mathcal{A}$ denote a monoidal $\mathcal{V}$ -category. We do not need $\mathcal{A}$ to be closed for the definition of Tambara module although we will require this restriction again later.

A *left Tambara module* on $\mathcal{A}$ is a $\mathcal{V}$ -functor $T : \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{V}$ together with a family of morphisms

$$\alpha_l(A, X, Y) : T(X, Y) \longrightarrow T(A \otimes X, A \otimes Y)$$

which are $\mathcal{V}$ -natural in each of the objects A , X and Y , satisfying the two equations $\alpha_l(I, X, Y) = 1_{T(X, Y)}$ and

$$\begin{array}{ccc} T(X, Y) & \xrightarrow{\alpha_l(A', X, Y)} & T(A' \otimes X, A' \otimes Y) \\ & \searrow \alpha_l(A \otimes A', X, Y) & \downarrow \alpha_l(A, A' \otimes X, A' \otimes Y) \\ & & T(A \otimes A' \otimes X, A \otimes A' \otimes Y). \end{array}$$

Similarly, a *right Tambara module* on $\mathcal{A}$ is a $\mathcal{V}$ -functor $T : \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{V}$ together with a family of morphisms

$$\alpha_r(B, X, Y) : T(X, Y) \longrightarrow T(X \otimes B, Y \otimes B)$$

which are $\mathcal{V}$ -natural in each of the objects B , X and Y , satisfying the two equations $\alpha_r(I, X, Y) = 1_{T(X, Y)}$ and

$$\begin{array}{ccc} T(X, Y) & \xrightarrow{\alpha_r(B, X, Y)} & T(X \otimes B, Y \otimes B) \\ & \searrow \alpha_r(B \otimes B', X, Y) & \downarrow \alpha_r(B', B \otimes X, B \otimes Y) \\ & & T(X \otimes B \otimes B', Y \otimes B \otimes B'). \end{array}$$

A *Tambara module* (T, α) on $\mathcal{A}$ is a $\mathcal{V}$ -functor $T : \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{V}$ together with both left and right Tambara module structures satisfying the “bimodule” compatibility condition

$$\begin{array}{ccc} T(X, Y) & \xrightarrow{\alpha_l(A, X, Y)} & T(A \otimes X, A \otimes Y) \\ \alpha_r(B, X, Y) \downarrow & & \downarrow \alpha_r(B, A \otimes X, A \otimes Y) \\ T(X \otimes B, Y \otimes B) & \xrightarrow{\alpha_l(A, X \otimes B, Y \otimes B)} & T(A \otimes X \otimes B, A \otimes Y \otimes B). \end{array}$$

The morphism defined to be the diagonal of the last square is denoted by

$$\alpha(A, B, X, Y) : T(X, Y) \longrightarrow T(A \otimes X \otimes B, A \otimes Y \otimes B)$$

and we can express a Tambara module structure purely in terms of this, however, we need to refer to the left and right structures below.

3.1. PROPOSITION. *Suppose $\mathcal{A}$ is a monoidal $\mathcal{V}$ -category and $T : \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{V}$ is a $\mathcal{V}$ -functor.*

- (a) *If $\mathcal{A}$ is right closed, there is a bijection between $\mathcal{V}$ -natural families of morphisms*

$$\alpha_l(A, X, Y) : T(X, Y) \longrightarrow T(A \otimes X, A \otimes Y)$$

and $\mathcal{V}$ -natural families of morphisms

$$\beta_l(A, X, Y) : T(X, Y^A) \longrightarrow T(A \otimes X, Y).$$

- (b) *Under the bijection of (a), the family α_l is a left Tambara structure if and only if the family β_l satisfies the two equations $\beta_l(I, X, Y) = 1_{T(X, Y)}$ and*

$$\begin{array}{ccc} T(X, Y^{A \otimes A'}) & \xrightarrow{\beta_l(A \otimes A', X, Y)} & T(A \otimes A' \otimes X, Y) \\ \downarrow = & & \uparrow \beta_l(A, A' \otimes X, Y) \\ T(X, (Y^A)^{A'}) & \xrightarrow{\beta_l(A', X, Y^A)} & T(A' \otimes X, Y^A). \end{array}$$

(c) If $\mathcal{A}$ is left closed, there is a bijection between $\mathcal{V}$ -natural families of morphisms

$$\alpha_r(B, X, Y) : T(X, Y) \longrightarrow T(X \otimes B, Y \otimes B)$$

and $\mathcal{V}$ -natural families of morphisms

$$\beta_r(B, X, Y) : T(X, {}^B Y) \longrightarrow T(X \otimes B, Y).$$

(d) Under the bijection of (c), the family α_r is a right Tambara structure if and only if the family β_r satisfies the two equations $\beta_r(I, X, Y) = 1_T(X, Y)$ and

$$\begin{array}{ccc} T(X, {}^{B \otimes B'} Y) & \xrightarrow{\beta_r(B \otimes B', X, Y)} & T(X \otimes B \otimes B', Y) \\ \downarrow = & & \uparrow \beta_r(B', X \otimes B, Y) \\ T(X, {}^B ({}^{B'} Y)) & \xrightarrow{\beta_r(B, X, {}^{B'} Y)} & T(X \otimes B, {}^{B'} Y). \end{array}$$

(e) If $\mathcal{A}$ is closed, the families α_l and α_r form a Tambara module structure if and only if the families β_l and β_r , corresponding under (a) and (c), satisfy the condition

$$\begin{array}{ccc} T(X, {}^B Y^A) & \xrightarrow{\beta_l(A, X, {}^B Y)} & T(A \otimes X, {}^B Y) \\ \beta_r(B, X, Y^A) \downarrow & & \downarrow \beta_r(B, A \otimes X, Y) \\ T(X \otimes B, Y^A) & \xrightarrow{\beta_l(A, X \otimes B, Y)} & T(A \otimes X \otimes B, Y). \end{array}$$

PROOF. The bijection of (a) is defined by the formulas

$$\beta_l(A, X, Y) = \left(T(X, Y^A) \xrightarrow{\alpha_l(A, X, Y^A)} T(A \otimes X, A \otimes Y^A) \xrightarrow{T(A \otimes X, e_r)} T(A \otimes X, Y) \right)$$

and

$$\alpha_l(A, X, Y) = \left(T(X, Y) \xrightarrow{T(X, d_r)} T(X, (A \otimes Y)^A) \xrightarrow{\beta_l(A, X, A \otimes Y)} T(A \otimes X, A \otimes Y) \right).$$

That the processes are mutually inverse uses the adjunction identities on the morphisms e and d . The bijection of (c) is obtained dually by reversing the tensor product. Translation of the conditions from the α to the β as required for (b), (d) and (e) is straightforward. ■

A left (respectively, right) Tambara module T on $\mathcal{A}$ will be called *strong* when the morphisms

$$\begin{aligned} \beta_l(A, X, Y) : T(X, Y^A) &\longrightarrow T(A \otimes X, Y) \\ (\text{respectively, } \beta_r(B, X, Y) : T(X, {}^B Y) &\longrightarrow T(X \otimes B, Y)) \end{aligned}$$

corresponding via Proposition 3.1 to the left (respectively, right) Tambara structure, are invertible. A Tambara module is called *left* (respectively, *right*) *strong* when it is strong as a left (respectively, right) module and *strong* when it is both left and right strong. In particular, notice that the hom $\mathcal{V}$ -functor (= identity module) of $\mathcal{A}$ is a strong Tambara module.

3.2. PROPOSITION. *Suppose $\mathcal{A}$ is a monoidal $\mathcal{V}$ -category and $T : \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{V}$ is a $\mathcal{V}$ -functor. If $\mathcal{A}$ is right (left) autonomous then every left (right) Tambara module is strong.*

PROOF. If A^* denotes a right dual for A with unit $\eta : I \longrightarrow A^* \otimes A$ then an inverse for β_l is defined by the composite

$$T(A \otimes X, Y) \xrightarrow{\alpha_l(A^*, A \otimes X, Y)} T(A^* \otimes A \otimes X, A^* \otimes Y) \xrightarrow{T(\eta, 1)} T(X, A^* \otimes Y) .$$

■

Write $\text{LTamb}(\mathcal{A})$ for the $\mathcal{V}$ -category whose objects are left Tambara modules $T = (T, \alpha_l)$ and whose hom $\text{LTamb}(\mathcal{A})(T, T')$ in $\mathcal{V}$ is defined to be the intersection over all A, X and Y of the equalizers of the pairs of morphisms:

$$[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}](T, T') \xrightarrow[\mathcal{V}(1, \alpha_l) \circ \text{pr}_{X, Y}]{\mathcal{V}(\alpha_l, 1) \circ \text{pr}_{A \otimes X, A \otimes Y}} \mathcal{V}(T(X, Y), T'(A \otimes X, A \otimes Y)) .$$

Equivalently, we can define the hom as an intersection of equalizers of pairs of morphisms:

$$[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}](T, T') \xrightarrow[\mathcal{V}(1, \beta_l) \circ \text{pr}_{X, Y^A}]{\mathcal{V}(\beta_l, 1) \circ \text{pr}_{A \otimes X, Y}} \mathcal{V}(T(X, Y^A), T'(A \otimes X, Y)) .$$

Composition is defined so that we have a $\mathcal{V}$ -functor $\iota : \text{LTamb}(\mathcal{A}) \longrightarrow [\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$ which forgets the left module structure on T . In fact, $\text{LTamb}(\mathcal{A})$ becomes a monoidal $\mathcal{V}$ -category in such a way that the forgetful $\mathcal{V}$ -functor ι becomes strong monoidal. For this, the monoidal structure on $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$ is the usual tensor product (= composition) of endomodules:

$$(T \otimes_{\mathcal{A}} T')(X, Y) = \int^Z T(X, Z) \otimes T'(Z, Y) .$$

When T and T' are left Tambara modules, the left Tambara structure

$$(T \otimes_{\mathcal{A}} T')(X, Y) \longrightarrow (T \otimes_{\mathcal{A}} T')(A \otimes X, A \otimes Y)$$

on $T \otimes_{\mathcal{A}} T'$ is defined by taking its composite with the coprojection copr_Z into the above coend to be the composite

$$\begin{aligned} T(X, Z) \otimes T'(Z, Y) &\xrightarrow{\alpha_l \otimes \alpha_l} T(A \otimes X, A \otimes Z) \otimes T'(A \otimes Z, A \otimes Y) \\ &\xrightarrow{\text{copr}_{A \otimes Z}} (T \otimes_{\mathcal{A}} T')(A \otimes X, A \otimes Y). \end{aligned}$$

Similarly we obtain monoidal $\mathcal{V}$ -categories $\text{RTamb}(\mathcal{A})$ and $\text{Tamb}(\mathcal{A})$ of right Tambara and all Tambara modules on $\mathcal{A}$.

We write $\text{LTamb}_s(\mathcal{A})$ for the full sub- $\mathcal{V}$ -category of $\text{LTamb}(\mathcal{A})$ consisting of the strong left Tambara modules. We write $\text{Tamb}_{ls}(\mathcal{A})$, $\text{Tamb}_{rs}(\mathcal{A})$ and $\text{Tamb}_s(\mathcal{A})$ for the full sub- $\mathcal{V}$ -categories of $\text{Tamb}(\mathcal{A})$ consisting of the left strong, right strong and strong Tambara modules respectively.

If $\mathcal{A}$ is autonomous then $\text{Tamb}(\mathcal{A}) = \text{Tamb}_{ls}(\mathcal{A}) = \text{Tamb}_{rs}(\mathcal{A}) = \text{Tamb}_s(\mathcal{A})$ by Proposition 3.2.

4. The Cayley functor

Consider a right closed monoidal $\mathcal{V}$ -category $\mathcal{A}$. There is a *Cayley $\mathcal{V}$ -functor*

$$\Upsilon : [\mathcal{A}, \mathcal{V}] \longrightarrow [\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$$

defined as follows. To each object $F \in [\mathcal{A}, \mathcal{V}]$, define $\Upsilon(F) = T_F$ by

$$T_F(X, Y) = F(Y^X).$$

The effect $\Upsilon_{F,G} : [\mathcal{A}, \mathcal{V}](F, G) \longrightarrow [\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}](T_F, T_G)$ of Υ on homs is defined by taking its composite with the projection

$$\text{pr}_{X,Y} : [\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}](T_F, T_G) \longrightarrow \mathcal{V}(F(Y^X), G(Y^X))$$

to be the projection

$$\text{pr}_{Y^X} : [\mathcal{A}, \mathcal{V}](F, G) \longrightarrow \mathcal{V}(F(Y^X), G(Y^X)).$$

4.1. PROPOSITION. *The Cayley $\mathcal{V}$ -functor Υ is strong monoidal; it takes Day convolution to composition of endomodules.*

PROOF. We have the calculation:

$$\begin{aligned}
(\Upsilon(F) \otimes_{\mathcal{A}} \Upsilon(G))(X, Y) &= \int^Z \Upsilon(F)(X, Z) \otimes \Upsilon(G)(Z, Y) \\
&= \int^Z F(Z^X) \otimes G(Y^Z) \\
&\cong \int^{Z, U, V} \mathcal{A}(U, Z^X) \otimes FU \otimes \mathcal{A}(V, Y^Z) \otimes GV \\
&\cong \int^{Z, U, V} \mathcal{A}(X \otimes U, Z) \otimes FU \otimes \mathcal{A}(Z \otimes V, Y) \otimes GV \\
&\cong \int^{U, V} \mathcal{A}(X \otimes U \otimes V, Y) \otimes FU \otimes GV \\
&\cong \int^{U, V} \mathcal{A}(U \otimes V, Y^X) \otimes FU \otimes GV \\
&\cong \Upsilon(F * G)(X, Y),
\end{aligned}$$

and of course $\Upsilon(\mathcal{A}(I, -))(X, Y) = \mathcal{A}(I, Y^X) \cong \mathcal{A}(X, Y)$. $\blacksquare$

In fact, Υ lands in the left Tambara modules by defining, for each $F \in [\mathcal{A}, \mathcal{V}]$, the structure

$$\alpha_l(A, X, Y) = \left(F(Y^X) \xrightarrow{F((d_r)^X)} F((A \otimes Y)^{A \otimes X}) \right)$$

on T_F . It is helpful to observe that the β_l corresponding to this α_l (via Proposition 3.1) is given by the identity

$$\beta_l(A, X, Y) = \left(F(Y^{A \otimes X}) \xrightarrow{1} F(Y^{A \otimes X}) \right),$$

showing that T_F becomes a strong left module. To see that there is a $\mathcal{V}$ -functor $\hat{\Upsilon} : [\mathcal{A}, \mathcal{V}] \longrightarrow \text{LTamb}_s(\mathcal{A})$ satisfying $\iota \circ \hat{\Upsilon} = \Upsilon$, we merely observe that

$$\text{pr}_{A \otimes X, Y} \circ \Upsilon_{F, G} = \text{pr}_{Y^{A \otimes X}} = \text{pr}_{(Y^A)^X} = \text{pr}_{X, Y^A} \circ \Upsilon_{F, G}.$$

4.2. PROPOSITION. *If $\mathcal{A}$ is a right closed monoidal $\mathcal{V}$ -category then the $\mathcal{V}$ -functor $\hat{\Upsilon} : [\mathcal{A}, \mathcal{V}] \longrightarrow \text{LTamb}_s(\mathcal{A})$ is an equivalence.*

PROOF. Define $\zeta : \text{LTamb}(\mathcal{A})(T_F, T_G) \longrightarrow [\mathcal{A}, \mathcal{V}](F, G)$ by $\text{pr}_Y \circ \zeta = \text{pr}_{I, Y} \circ \iota_{T_F, T_G}$. Then

$$\text{pr}_Y \circ \zeta \circ \hat{\Upsilon}_{F, G} = \text{pr}_{I, Y} \circ \iota_{T_F, T_G} \circ \hat{\Upsilon}_{F, G} = \text{pr}_{I, Y} \circ \Upsilon_{F, G} = \text{pr}_Y$$

and

$$\begin{aligned}
\text{pr}_{X, Y} \circ \iota_{T_F, T_G} \circ \hat{\Upsilon}_{F, G} \circ \zeta &= \text{pr}_{X, Y} \circ \Upsilon_{F, G} \circ \zeta \\
&= \text{pr}_{Y^X} \circ \zeta \\
&= \text{pr}_{I, Y^X} \circ \iota_{T_F, T_G} \\
&= \text{pr}_{X, Y} \circ \iota_{T_F, T_G}.
\end{aligned}$$

It follows that ζ is the inverse of $\hat{\Upsilon}_{F,G}$, so that $\hat{\Upsilon}$ is fully faithful. To see that $\hat{\Upsilon}$ is essentially surjective on objects, take a strong left module T . Put $FY = T(I, Y)$ as a $\mathcal{V}$ -functor in Y . Then the isomorphism $\beta_l(X, I, Y)$ yields

$$T_F(X, Y) = F(Y^X) = T(I, Y^X) \cong T(X, Y);$$

so $\hat{\Upsilon}(F) \cong T$. ■

Now suppose we have an object (F, θ) of the lax centre $\mathcal{Z}_l[\mathcal{A}, \mathcal{V}]$ of $[\mathcal{A}, \mathcal{V}]$. Then T_F becomes a right Tambara module by defining

$$\alpha_r(B, X, Y) = \left(F(Y^X) \xrightarrow{F((d_l)^X)} F(B(Y \otimes B)^X) \xrightarrow{\theta_{B, (Y \otimes B)^X}} F(Y \otimes B)^{X \otimes B} \right).$$

If $\mathcal{A}$ is left closed, the β_r corresponding to this α_r (via Proposition 3.1) is defined by

$$\beta_r(B, X, Y) = \left(F(BY^X) \xrightarrow{\theta_{B, Y^X}} F(Y^{X \otimes B}) \right).$$

It is easy to see that, in this way, $T_F = \hat{\Upsilon}(F)$ actually becomes a (two-sided) Tambara module which we write as $\hat{\Upsilon}(F, \theta)$, and we have a $\mathcal{V}$ -functor

$$\hat{\Upsilon} : \mathcal{Z}_l[\mathcal{A}, \mathcal{V}] \longrightarrow \text{Tamb}_{ls}(\mathcal{A}).$$

4.3. PROPOSITION. *If $\mathcal{A}$ is a closed monoidal $\mathcal{V}$ -category then the $\mathcal{V}$ -functor*

$$\hat{\Upsilon} : \mathcal{Z}_l[\mathcal{A}, \mathcal{V}] \longrightarrow \text{Tamb}_{ls}(\mathcal{A})$$

is an equivalence which restricts to an equivalence

$$\hat{\Upsilon} : \mathcal{Z}[\mathcal{A}, \mathcal{V}] \longrightarrow \text{Tamb}_s(\mathcal{A}).$$

PROOF. The proof of full faithfulness proceeds along the lines of the beginning of the proof of Proposition 4.2. For essential surjectivity on objects, take a left strong Tambara module (T, α) . Then $\beta_l(A, X, Y) : T(X, Y^A) \longrightarrow T(A \otimes X, Y)$ is invertible. Define the $\mathcal{V}$ -functor $F : \mathcal{A} \longrightarrow \mathcal{V}$ by $FX = T(I, X)$ as in the proof of Proposition 4.2, and define $\theta_{A, Y} : F(^AY) \longrightarrow F(Y^A)$ to be the composite

$$T(I, ^AY) \xrightarrow{\beta_r(A, I, Y)} T(A, Y) \xrightarrow{\beta_l(A, I, Y)^{-1}} T(I, Y^A).$$

This is easily seen to yield an object (F, θ) of the lax centre $\mathcal{Z}_l[\mathcal{A}, \mathcal{V}]$ with $\hat{\Upsilon}(F, \theta) \cong T_F$. Thus we have the first equivalence. Clearly θ is invertible if and only if β_r is; the second equivalence follows. ■

5. The double monad

Tambara modules are actually Eilenberg-Moore coalgebras for a fairly obvious comonad on $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$. We begin by looking at the case of left modules.

Let $\Theta_l : [\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}] \longrightarrow [\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$ be the $\mathcal{V}$ -functor defined by the end

$$\Theta_l(T)(X, Y) = \int_A T(A \otimes X, A \otimes Y).$$

There is a $\mathcal{V}$ -natural family $\epsilon_T : \Theta_l(T) \longrightarrow T$ defined by the projections

$$\text{pr}_I : \int_A T(A \otimes X, A \otimes Y) \longrightarrow T(X, Y).$$

There is a $\mathcal{V}$ -natural family $\delta_T : \Theta_l(T) \longrightarrow \Theta_l(\Theta_l(T))$ defined by taking its composite with the projection

$$\text{pr}_{B,C} : \int_{B,C} T(B \otimes C \otimes X, B \otimes C \otimes Y) \longrightarrow T(B \otimes C \otimes X, B \otimes C \otimes Y)$$

to be the projection

$$\text{pr}_{B \otimes C} : \int_A T(A \otimes X, A \otimes Y) \longrightarrow T(B \otimes C \otimes X, B \otimes C \otimes Y).$$

It is now easily checked that $\Theta_l = (\Theta_l, \delta, \epsilon)$ is a comonad on $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$.

There are also a comonad Θ_r on $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$, a distributive law $\Theta_r \Theta_l \cong \Theta_l \Theta_r$, and a comonad $\Theta = \Theta_r \Theta_l$:

$$\Theta_r(T)(X, Y) = \int_B T(X \otimes B, Y \otimes B)$$

and

$$\Theta(T)(X, Y) = \int_{A,B} T(A \otimes X \otimes B, A \otimes Y \otimes B).$$

We can easily identify the $\mathcal{V}$ -categories of Eilenberg-Moore coalgebras for these three comonads.

5.1. PROPOSITION. *There are isomorphisms of $\mathcal{V}$ -categories*

- $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]^{\Theta_l} \cong \text{LTamb}(\mathcal{A}),$
- $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]^{\Theta_r} \cong \text{RTamb}(\mathcal{A}),$ and
- $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]^{\Theta} \cong \text{Tamb}(\mathcal{A}).$

In fact, Θ_l , Θ_r and Θ are all monoidal comonads on $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]$. For example, the structure on Θ_l is provided by the $\mathcal{V}$ -natural transformations $\Theta_l(T) \otimes_{\mathcal{A}} \Theta_l(T') \longrightarrow \Theta_l(T \otimes_{\mathcal{A}} T')$ and $\mathcal{A}(-, -) \longrightarrow \Theta_l(\mathcal{A}(-, -))$ with components

$$\int_A^Z T(A \otimes X, A \otimes Z) \otimes \int_B T'(B \otimes X, B \otimes Z) \longrightarrow \int_C \int^U T(C \otimes X, U) \otimes T'(U, C \otimes Y) \quad (1)$$

and

$$\mathcal{A}(X, Y) \longrightarrow \int_A \mathcal{A}(A \otimes X, A \otimes Y) \quad (2)$$

defined as follows. The morphism (1) is determined by its precomposite with the coprojection copr_Z and postcomposite with the projection pr_C ; the result is defined to be the composite

$$\begin{aligned} \int_A T(A \otimes X, A \otimes Z) \otimes \int_B T'(B \otimes X, B \otimes Z) \\ \xrightarrow{\text{pr}_C \otimes \text{pr}_C} T(C \otimes X, C \otimes Z) \otimes T'(C \otimes Z, C \otimes Y) \\ \xrightarrow{\text{copr}_{C \otimes Z}} \int^U T(C \otimes X, U) \otimes T'(U, C \otimes Y). \end{aligned}$$

The morphism (2) is simply the coprojection copr_I . It follows that $[\mathcal{A}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]^{\Theta_l}$ becomes monoidal with the underlying functor becoming strong monoidal; see [Moe02] and [McC02]. Clearly we have:

5.2. PROPOSITION. *The isomorphisms of Proposition 5.1 are monoidal.*

The next thing to observe is that Θ_l , Θ_r and Θ all have left adjoints Φ_l , Φ_r and Φ which therefore become opmonoidal monads whose $\mathcal{V}$ -categories of Eilenberg-Moore algebras are monoidally isomorphic to $\text{LTamb}(\mathcal{A})$, $\text{RTamb}(\mathcal{A})$ and $\text{Tamb}(\mathcal{A})$, respectively. Straightforward applications of the Yoneda Lemma, show that the formulas for these adjoints are

$$\begin{aligned} \Phi_l(S)(U, V) &= \int^{A, X, Y} \mathcal{A}(U, A \otimes X) \otimes \mathcal{A}(A \otimes Y, V) \otimes S(X, Y), \\ \Phi_r(S)(U, V) &= \int^{B, X, Y} \mathcal{A}(U, X \otimes B) \otimes \mathcal{A}(Y \otimes B, V) \otimes S(X, Y), \quad \text{and} \\ \Phi(S)(U, V) &= \int^{A, B, X, Y} \mathcal{A}(U, A \otimes X \otimes B) \otimes \mathcal{A}(A \otimes Y \otimes B, V) \otimes S(X, Y). \end{aligned}$$

Recall that left adjoint $\mathcal{V}$ -functors $\Psi : [\mathcal{X}^{\text{op}}, \mathcal{V}] \longrightarrow [\mathcal{Y}^{\text{op}}, \mathcal{V}]$ are equivalent to $\mathcal{V}$ -functors $\check{\Psi} : \mathcal{Y}^{\text{op}} \otimes \mathcal{X} \longrightarrow \mathcal{V}$, which are also called modules $\check{\Psi} : \mathcal{X} \dashrightarrow \mathcal{Y}$ from $\mathcal{X}$ to $\mathcal{Y}$. The equivalence is defined by:

$$\check{\Psi}(Y, X) = \Psi(\mathcal{X}(-, X))(Y)$$

and

$$\Psi(M)(Y) = \int^X \check{\Psi}(Y, X) \otimes M(X).$$

It follows that Φ_l , Φ_r and Φ determine monads $\check{\Phi}_l$, $\check{\Phi}_r$ and $\check{\Phi}$ on $\mathcal{A}^{\text{op}} \otimes \mathcal{A}$ in the bicategory $\mathcal{V}\text{-}\mathbf{Mod}$. The formulas are:

$$\begin{aligned} \check{\Phi}_l(X, Y, U, V) &= \int^A \mathcal{A}(U, A \otimes X) \otimes \mathcal{A}(A \otimes Y, V), \\ \check{\Phi}_r(X, Y, U, V) &= \int^B \mathcal{A}(U, X \otimes B) \otimes \mathcal{A}(Y \otimes B, V), \quad \text{and} \\ \check{\Phi}(X, Y, U, V) &= \int^{A,B} \mathcal{A}(U, A \otimes X \otimes B) \otimes \mathcal{A}(A \otimes Y \otimes B, V). \end{aligned}$$

6. Doubles

The bicategory $\mathcal{V}\text{-}\mathbf{Mod}$ admits the Kleisli construction for monads. Write $\mathcal{D}_l\mathcal{A}$, $\mathcal{D}_r\mathcal{A}$ and $\mathcal{D}\mathcal{A}$ for the Kleisli $\mathcal{V}$ -categories for the monads $\check{\Phi}_l$, $\check{\Phi}_r$ and $\check{\Phi}$ on $\mathcal{A}^{\text{op}} \otimes \mathcal{A}$ in the bicategory $\mathcal{V}\text{-}\mathbf{Mod}$. We call them the *left double*, *right double* and *double* of the monoidal $\mathcal{V}$ -category $\mathcal{A}$. They all have the same objects as $\mathcal{A}^{\text{op}} \otimes \mathcal{A}$. The homs are defined by

$$\begin{aligned} \mathcal{D}_l\mathcal{A}((X, Y), (U, V)) &= \int^A \mathcal{A}(U, A \otimes X) \otimes \mathcal{A}(A \otimes Y, V), \\ \mathcal{D}_r\mathcal{A}((X, Y), (U, V)) &= \int^B \mathcal{A}(U, X \otimes B) \otimes \mathcal{A}(Y \otimes B, V), \quad \text{and} \\ \mathcal{D}\mathcal{A}((X, Y), (U, V)) &= \int^{A,B} \mathcal{A}(U, A \otimes X \otimes B) \otimes \mathcal{A}(A \otimes Y \otimes B, V). \end{aligned}$$

6.1. PROPOSITION. *There are canonical equivalences of $\mathcal{V}$ -categories:*

- $\Xi_l : \text{LTamb}(\mathcal{A}) \simeq [\mathcal{D}_l\mathcal{A}, \mathcal{V}]$,
- $\Xi_r : \text{RTamb}(\mathcal{A}) \simeq [\mathcal{D}_r\mathcal{A}, \mathcal{V}]$, and
- $\Xi : \text{Tamb}(\mathcal{A}) \simeq [\mathcal{D}\mathcal{A}, \mathcal{V}]$.

It follows from the main result of Day [Day70] that these doubles $\mathcal{D}_l\mathcal{A}$, $\mathcal{D}_r\mathcal{A}$ and $\mathcal{D}\mathcal{A}$ all admit promonoidal structures (P_l, J_l) , (P_r, J_r) and (P, J) for which the equivalences in Proposition 6.1 become monoidal when the right-hand sides are given the corresponding convolution structures. For example, we calculate that P_l and J_l are as follows:

$$\begin{aligned} P_l((X, Y), (U, V); (H, K)) &= (\mathcal{D}_l\mathcal{A}((X, Y), -) \otimes_{\mathcal{A}} \mathcal{D}_l\mathcal{A}((U, V), -))(H, K) \\ &= \int^{Z, A, B} \mathcal{A}(H, A \otimes X) \otimes \mathcal{A}(A \otimes Y, Z) \otimes \mathcal{A}(Z, B \otimes U) \otimes \mathcal{A}(B \otimes V, K) \\ &= \int^{A, B} \mathcal{A}(H, A \otimes X) \otimes \mathcal{A}(A \otimes Y, B \otimes U) \otimes \mathcal{A}(B \otimes V, K) \end{aligned}$$

and

$$J_l(H, K) = \mathcal{A}(H, K).$$

Furthermore, there are some special morphisms that exist in these doubles $\mathcal{D}_l\mathcal{A}$, $\mathcal{D}_r\mathcal{A}$ and $\mathcal{D}\mathcal{A}$. Let $\tilde{\alpha}_l : (X, Y) \longrightarrow (A \otimes X, A \otimes Y)$ denote the morphism in $\mathcal{D}_l\mathcal{A}$ defined by the composite

$$\begin{aligned} I &\xrightarrow{j_{A \otimes X} \otimes j_{A \otimes Y}} \mathcal{A}(A \otimes X, A \otimes X) \otimes \mathcal{A}(A \otimes Y, A \otimes Y) \\ &\xrightarrow{\text{copr}_A} \mathcal{D}_l\mathcal{A}((X, Y), (A \otimes X, A \otimes Y)). \end{aligned}$$

The $\mathcal{V}$ -functor Ξ_l has the property that $\Xi_l(T, \alpha_l)(X, Y) = T(X, Y)$ and $\Xi_l(T, \alpha_l)(\tilde{\alpha}_l) = \alpha_l$. When $\mathcal{A}$ is right closed, we let $\tilde{\beta}_l : (X, Y^A) \longrightarrow (A \otimes X, Y)$ denote the morphism in $\mathcal{D}_l\mathcal{A}$ defined by the composite

$$I \xrightarrow{j_{A \otimes X} \otimes e_r} \mathcal{A}(A \otimes X, A \otimes X) \otimes \mathcal{A}(A \otimes Y^A, Y) \xrightarrow{\text{copr}_A} \mathcal{D}_l\mathcal{A}((X, Y^A), (A \otimes X, Y)).$$

Then $\Xi_l(T, \alpha_l)(\tilde{\beta}_l) = \beta_l$.

Similarly, we have the morphism $\tilde{\alpha}_r : (X, Y) \longrightarrow (X \otimes B, Y \otimes B)$ in $\mathcal{D}_r\mathcal{A}$, and also, when $\mathcal{A}$ is left closed, the morphism $\tilde{\beta}_r : (X, {}^B Y) \longrightarrow (X \otimes B, Y)$.

There are $\mathcal{V}$ -functors $\mathcal{D}_l\mathcal{A} \longrightarrow \mathcal{D}\mathcal{A} \longleftarrow \mathcal{D}_r\mathcal{A}$ which are the identity functions on objects and are defined on homs using projections with $B = I$ for the left leg and the projections $A = I$ for the second leg. In this way, the morphisms $\tilde{\alpha}_l$ and $\tilde{\alpha}_r$ can be regarded also as morphisms of $\mathcal{D}\mathcal{A}$. Under closedness assumptions, the morphisms $\tilde{\beta}_l$ and $\tilde{\beta}_r$ can also be regarded as morphisms of $\mathcal{D}\mathcal{A}$.

Let Σ_l denote the set of morphisms $\tilde{\beta}_l : (X, Y^A) \longrightarrow (A \otimes X, Y)$, let Σ_r denote the set of morphisms $\tilde{\beta}_r : (X, {}^B Y) \longrightarrow (X \otimes B, Y)$, and let Σ denote the set of morphisms $\Sigma = \Sigma_l \cup \Sigma_r$. Under appropriate closedness assumptions on $\mathcal{A}$, we can form various $\mathcal{V}$ -categories of fractions such as:

- $\text{L}\mathcal{D}\mathcal{A} = \mathcal{D}_l\mathcal{A}[\Sigma_l^{-1}]$ and $\text{R}\mathcal{D}\mathcal{A} = \mathcal{D}_r\mathcal{A}[\Sigma_r^{-1}]$,
- $\mathcal{D}_{ls}\mathcal{A} = \mathcal{D}\mathcal{A}[\Sigma_l^{-1}]$ and $\mathcal{D}_{rs}\mathcal{A} = \mathcal{D}\mathcal{A}[\Sigma_r^{-1}]$, and
- $\mathcal{D}_s\mathcal{A} = \mathcal{D}\mathcal{A}[\Sigma^{-1}]$.

The following result is now automatic.

6.2. THEOREM. *For a closed monoidal $\mathcal{V}$ -category $\mathcal{A}$, there are equivalences of $\mathcal{V}$ -categories:*

- $[\text{L}\mathcal{D}\mathcal{A}, \mathcal{V}] \simeq \text{LTamb}_s(\mathcal{A}) \simeq [\mathcal{A}, \mathcal{V}]$,
- $[\mathcal{D}_{ls}\mathcal{A}, \mathcal{V}] \simeq \text{Tamb}_{ls}(\mathcal{A}) \simeq \mathcal{Z}_l[\mathcal{A}, \mathcal{V}]$, and
- $[\mathcal{D}_s\mathcal{A}, \mathcal{V}] \simeq \text{Tamb}_s(\mathcal{A}) \simeq \mathcal{Z}[\mathcal{A}, \mathcal{V}]$.

The first equivalence of Theorem 6.2 implies that $L\mathcal{D}\mathcal{A}$ and $\mathcal{A}$ are Morita equivalent. This begs the question of whether there is a $\mathcal{V}$ -functor relating them more directly. Indeed there is. We have a $\mathcal{V}$ -functor

$$\Pi : \mathcal{D}_l\mathcal{A} \longrightarrow \mathcal{A}$$

defined on objects by $\Pi(X, Y) = Y^X$ and by defining the effect

$$\Pi : \mathcal{D}_l\mathcal{A}((X, Y), (U, V)) \longrightarrow \mathcal{A}(Y^X, V^U)$$

on hom objects to have its composite with the A -coprojection equal to the composite

$$\begin{aligned} & \mathcal{A}(U, A \otimes X) \otimes \mathcal{A}(A \otimes Y, V) \\ & \xrightarrow{V(-) \otimes (-)^{A \otimes X}} \mathcal{A}(V^{A \otimes X}, V^U) \otimes \mathcal{A}((A \otimes Y)^{A \otimes X}, V^{A \otimes X}) \\ & \xrightarrow{\text{composition}} \mathcal{A}((A \otimes Y)^{A \otimes X}, V^U) \\ & \xrightarrow{\mathcal{A}((d_r)^X, V^U)} \mathcal{A}(Y^X, V^U). \end{aligned}$$

It is easy to see that Π takes the morphisms $\tilde{\beta}_l : (X, Y^A) \longrightarrow (A \otimes X, Y)$ to isomorphisms. So Π induces a $\mathcal{V}$ -functor

$$\hat{\Pi} : L\mathcal{D}_l\mathcal{A} \longrightarrow \mathcal{A};$$

this induces the first equivalence of Theorem 6.2.

For closed monoidal $\mathcal{A}$, the second and third equivalences of Theorem 6.2 show that both the lax centre and the centre of the convolution monoidal $\mathcal{V}$ -category $[\mathcal{A}, \mathcal{V}]$ are again functor $\mathcal{V}$ -categories $[\mathcal{D}_{ls}\mathcal{A}, \mathcal{V}]$ and $[\mathcal{D}_s\mathcal{A}, \mathcal{V}]$. Since $\mathcal{Z}_l[\mathcal{A}, \mathcal{V}]$ and $\mathcal{Z}[\mathcal{A}, \mathcal{V}]$ are monoidal with the tensor products colimit preserving in each variable, using the correspondence in [Day70], there are lax braided and braided promonoidal structures on $\mathcal{D}_{ls}\mathcal{A}$ and $\mathcal{D}_s\mathcal{A}$ which are such that $[\mathcal{D}_{ls}\mathcal{A}, \mathcal{V}]$ and $[\mathcal{D}_s\mathcal{A}, \mathcal{V}]$ become closed monoidal under convolution, and the equivalences of Theorem 6.2 become lax braided and braided monoidal equivalences.

6.3. REMARK.

- We are grateful to Brian Day for pointing out that the $\mathcal{V}$ -category $\mathcal{A}_M$ appearing in [DS07] is equivalent to the full sub- $\mathcal{V}$ -category of $\mathcal{D}\mathcal{A}$ consisting of the objects of the form (I, Y) .
- He also pointed out that a consequence of Theorem 6.2 is that the centre of $\mathcal{V}$ as a $\mathcal{V}$ -category is equivalent to $\mathcal{V}$ itself. This also can be seen directly by using the $\mathcal{V}$ -naturality in X of the centre structure $u_X : A \otimes X \longrightarrow X \otimes A$ on an object A of $\mathcal{V}$, and the fact that $u_I = 1$, to deduce that $u_X = c_{A, X}$ (the symmetry of $\mathcal{V}$). Generally, the centre of $\mathcal{V}$ as a monoidal **Set**-category is not equivalent to $\mathcal{V}$.

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